

The Three Aroma Types

Primary · Secondary · Tertiary

PRIMARY AROMAS

Where they come from

Primary aromas come directly from the grape itself — the fruit, the variety, and the place it was grown. Nothing has been done to the wine yet. These are the raw aromatic compounds present in the grape berry, shaped entirely by nature.

How they form

Terpenes, thiols, and pyrazines develop in the grape skin and pulp during the growing season. Temperature, sunlight, and soil all influence which compounds develop and how intensely. This is why the same grape variety smells different depending on where it's grown.

Fruit	Anything that smells like fresh fruit — berries, citrus, stone fruit, tropical fruit. The ripeness level signals climate: red/dark fruit = warmer; green/citrus = cooler.
Floral	Rose, violet, honeysuckle, jasmine. Common in aromatic varieties like Viognier, Gewurztraminer, and Nebbiolo.
Herbaceous	Green pepper, grass, tomato leaf, asparagus. Caused by pyrazine compounds in the skin — most prominent in cooler climates or less ripe vintages.
Earth / Mineral	Flint, wet stones, chalk, volcanic ash, saline. Linked to soil composition and vineyard environment rather than the grape variety itself.
Spice	Black/white pepper (rotundone compound), licorice. These are grape-derived, not from oak — Syrah's black pepper is a classic example.

Quick test

If you could smell a freshly crushed grape and recognize the aroma — it's primary.

SECONDARY AROMAS

Where they come from

Secondary aromas come from fermentation — specifically from yeast activity converting grape sugar into alcohol. The winemaker's choices at this stage have a huge impact on the final aroma profile.

How they form

Yeast produces esters, fatty acids, and other aromatic compounds as a byproduct of fermentation. The yeast strain chosen, fermentation temperature, and whether the wine undergoes malolactic fermentation (MLF) all determine what secondary aromas develop.

Butter / Cream	Produced by malolactic fermentation (MLF) — where sharp malic acid (green apple) is converted by bacteria into softer lactic acid (milk/cream). Classic in oaked Chardonnay.
Bread / Brioche	Yeast autolysis — when wine sits on dead yeast cells (lees). Classic in Champagne and traditional method sparkling wines. Also appears as toast or biscuit.
Yogurt / Cheese	Byproduct of MLF and lactic acid bacteria activity. Adds a creamy, tangy quality to the mid-palate.
Crème Brûlée	Combination of lactic richness from MLF and slight caramelisation from fermentation temperature. Often found in full-bodied whites.
Nutty	Can develop from oxidative handling during fermentation or from extended lees contact. Common in some white Burgundy and Champagne styles.

Quick test

If it smells like a dairy or a bakery — it's probably secondary.

TERTIARY AROMAS (The Bouquet)

Where they come from

Tertiary aromas develop after fermentation through aging — either in oak barrels or in the bottle over time. Once a wine develops tertiary aromas it stops being called 'aromas' and is technically referred to as the bouquet.

How they form — two pathways:

Oak aging:	The barrel itself imparts compounds. New French oak gives vanilla, clove, and subtle smoke. American oak gives stronger vanilla, coconut, and dill. The toasting level of the barrel matters — lightly toasted gives wood spice, heavily toasted gives coffee, chocolate, and smoke. Oxygen slowly entering through the wood also softens tannins and rounds the wine out.
Bottle aging (reductive):	With no new oxygen, the wine transforms slowly through chemical reactions between acids, alcohols, and other compounds. This is where the most complex and fascinating aromas develop.

Leather / Tobacco	Develops in red wines through long bottle aging. Tannins polymerize and the fruit recedes, allowing these savoury, complex notes to emerge.
Forest Floor / Earth	Mushroom, truffle, wet leaves, damp soil. Common in aged Pinot Noir and Nebbiolo. A sign of genuine bottle development.
Dried Fruit	Figs, dried cherry, prune, leather. Fruit character from primary aromas transforms into richer, more concentrated dried versions over time.
Vanilla / Coconut	Oak-derived. Vanillin from French oak, coconut (lactones) from American oak. More prominent in younger oak-aged wines.
Coffee / Chocolate	From heavily toasted barrels. Common in big reds aged in new oak — Napa Cabernet, Barossa Shiraz.
Petrol / Gasoline	TDN compound — develops in aged Riesling. Considered a mark of quality in great Mosel or Clare Valley examples.

Honey / Beeswax	Develops in aged white wines, particularly Chenin Blanc and white Burgundy. The primary fruit slowly gives way to these richer, waxy qualities.
Smoke / Toast	From oak stave toasting. Also found in wines from volcanic soils (Assyrtiko) where it can be a primary-tertiary crossover.

Reading All Three Layers

When you put your nose in a glass you're smelling all three layers at once. Training yourself to separate them tells you a lot about the wine — how ripe the vintage was (primary), how the winemaker handled fermentation (secondary), and how the wine was aged and for how long (tertiary). It's essentially reading the wine's entire biography through smell.

MLF — The Key Secondary Process to Know

Malolactic fermentation is the single biggest winemaker decision affecting secondary aromas. Full MLF = creamy, buttery, round. No MLF = crisper, more acidic, fruit-forward. Most red wines go through full MLF. For whites it's a choice — Chablis producers often block it to preserve freshness, while California Chardonnay producers embrace it for richness.